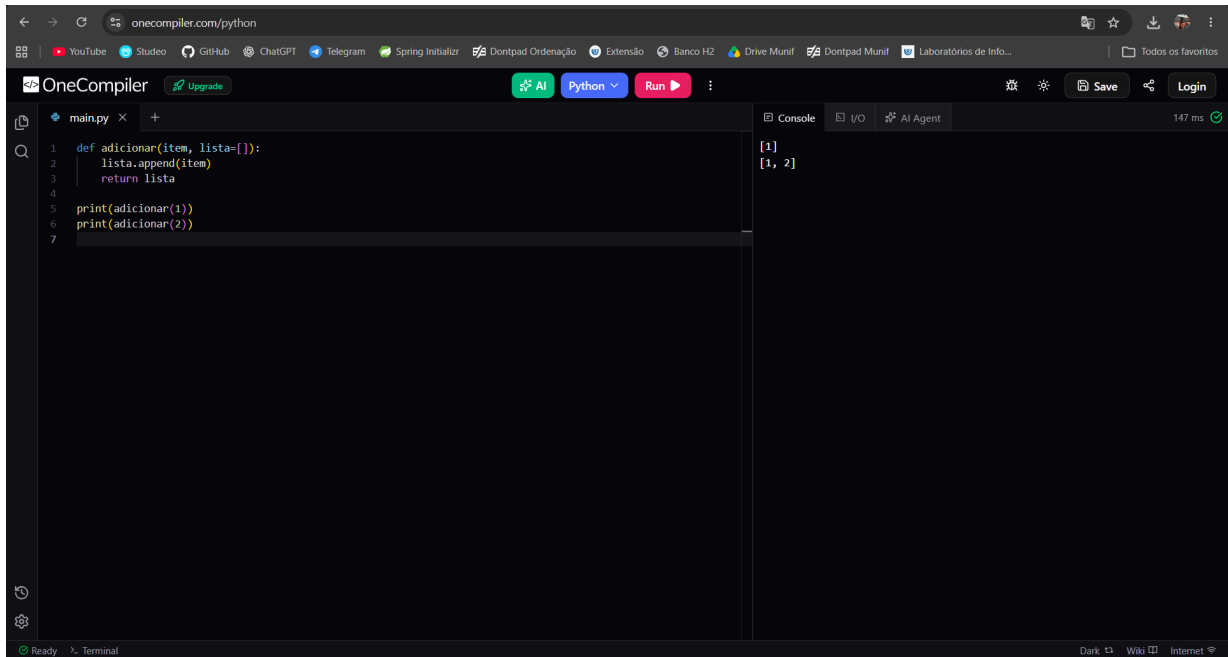


Aula 008 - Subprogramas

Atividade 1 - Python



The screenshot shows the OneCompiler Python IDE interface. The editor displays a Python script named 'main.py' with the following code:

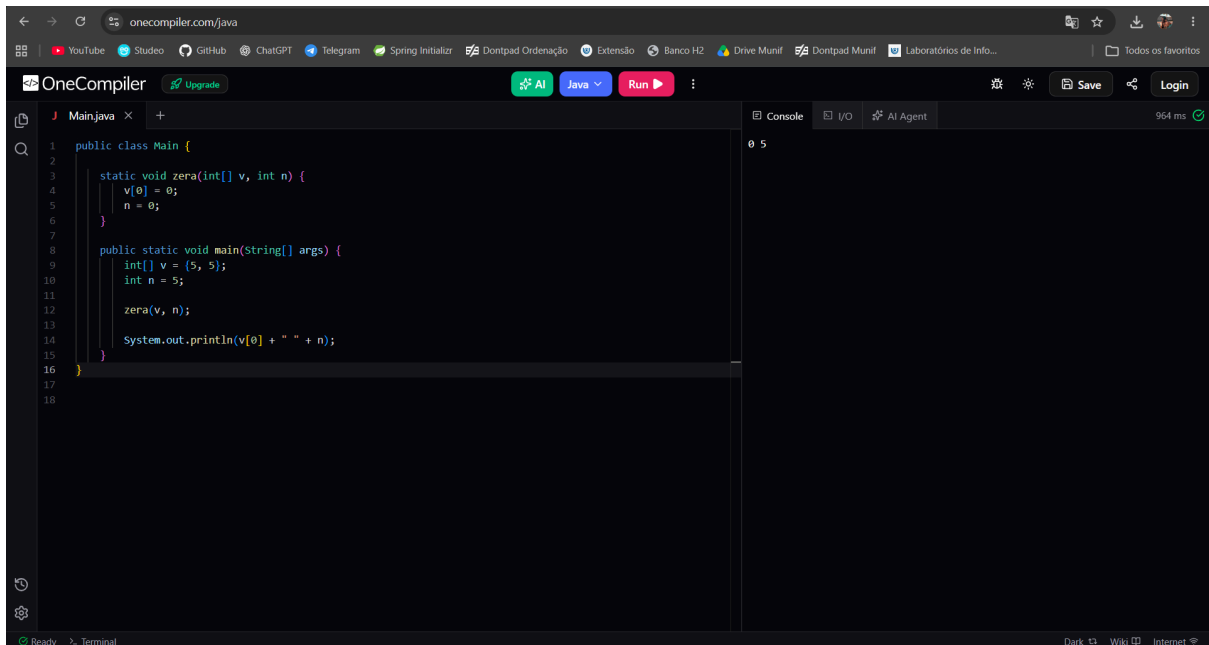
```
1 def adicionar(item, lista=[]):  
2     lista.append(item)  
3     return lista  
4  
5 print(adicionar(1))  
6 print(adicionar(2))  
7
```

The console on the right shows the output of the program:

```
[1]  
[1, 2]
```

The execution time is 147 ms. The interface includes a top bar with navigation icons, a search bar, and a bottom status bar showing 'Ready' and 'Terminal'.

Atividade 2 - Java



The screenshot shows the OneCompiler Java IDE interface. The editor displays a Java class named 'Main.java' with the following code:

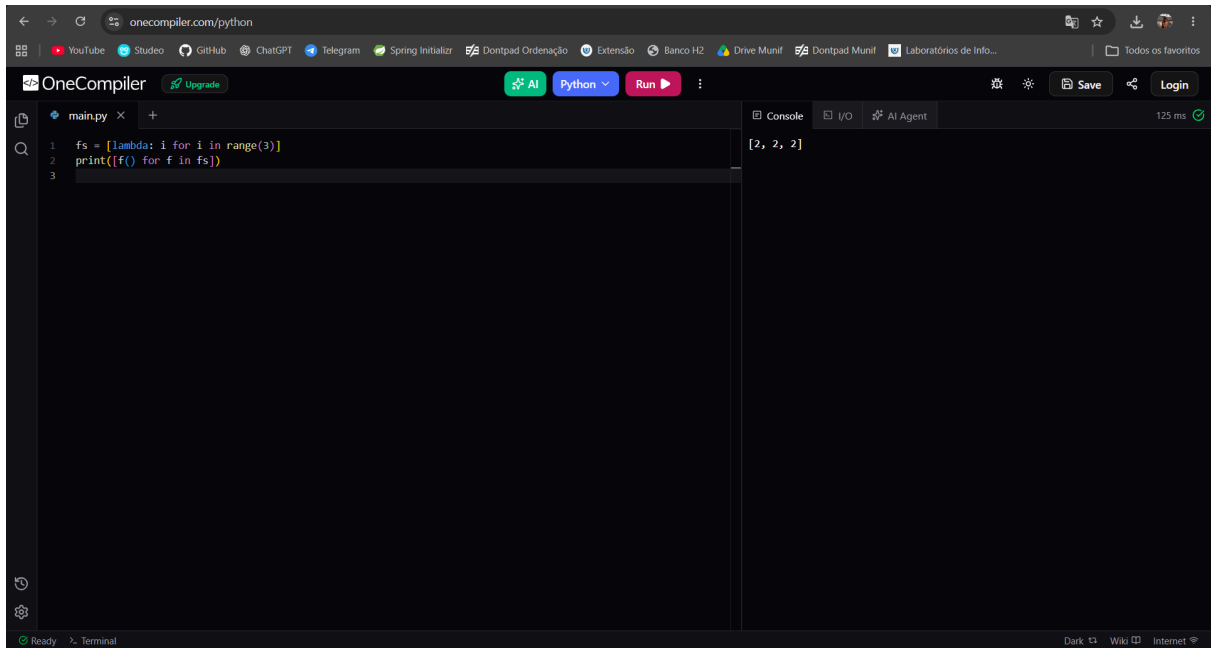
```
1 public class Main {  
2  
3     static void zera(int[] v, int n) {  
4         v[0] = 0;  
5         n = 0;  
6     }  
7  
8     public static void main(String[] args) {  
9         int[] v = {5, 5};  
10        int n = 5;  
11  
12        zera(v, n);  
13  
14        System.out.println(v[0] + " " + n);  
15    }  
16 }  
17  
18
```

The console on the right shows the output of the program:

```
0 5
```

The execution time is 964 ms. The interface includes a top bar with navigation icons, a search bar, and a bottom status bar showing 'Ready' and 'Terminal'.

Atividade 3 - Python

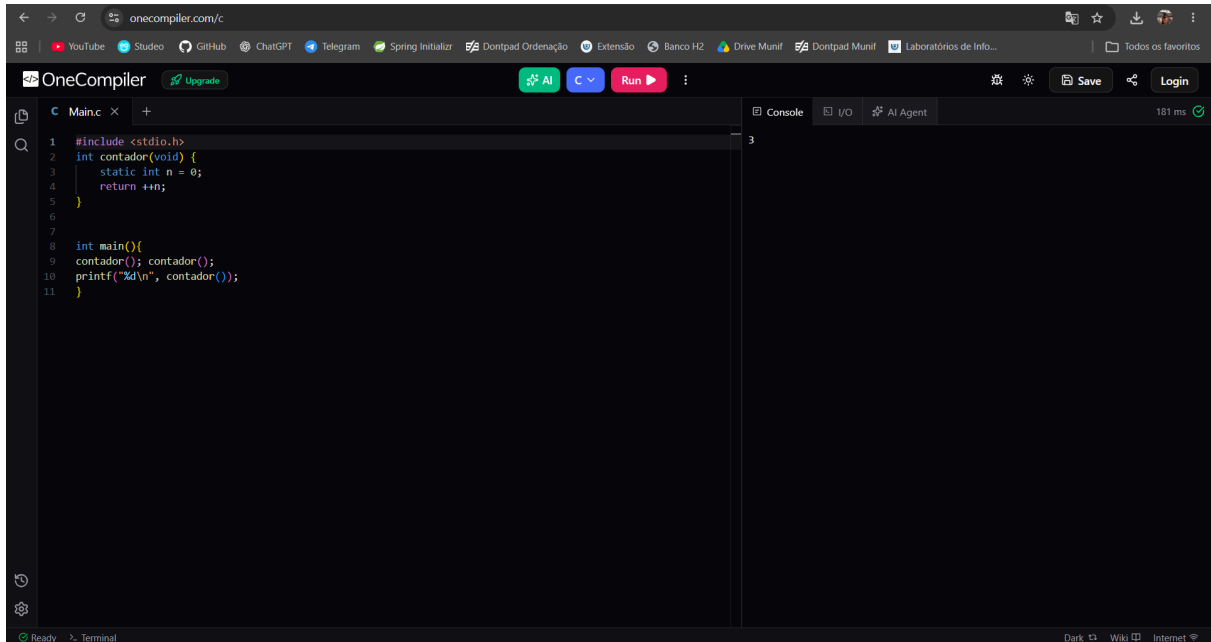


The screenshot shows the OneCompiler Python IDE interface. The editor displays a Python script in a file named `main.py`. The code defines a lambda function `fs` that takes `i` as an argument and returns `i` for `i` in the range of 3. It then prints the result of applying this function to each element `f` in the list `fs`.

```
1 fs = [lambda: i for i in range(3)]
2 print([f() for f in fs])
3
```

The console on the right shows the output of the program: `[2, 2, 2]`. The status bar at the bottom indicates the program is ready and the terminal is active.

Atividade 4 - C

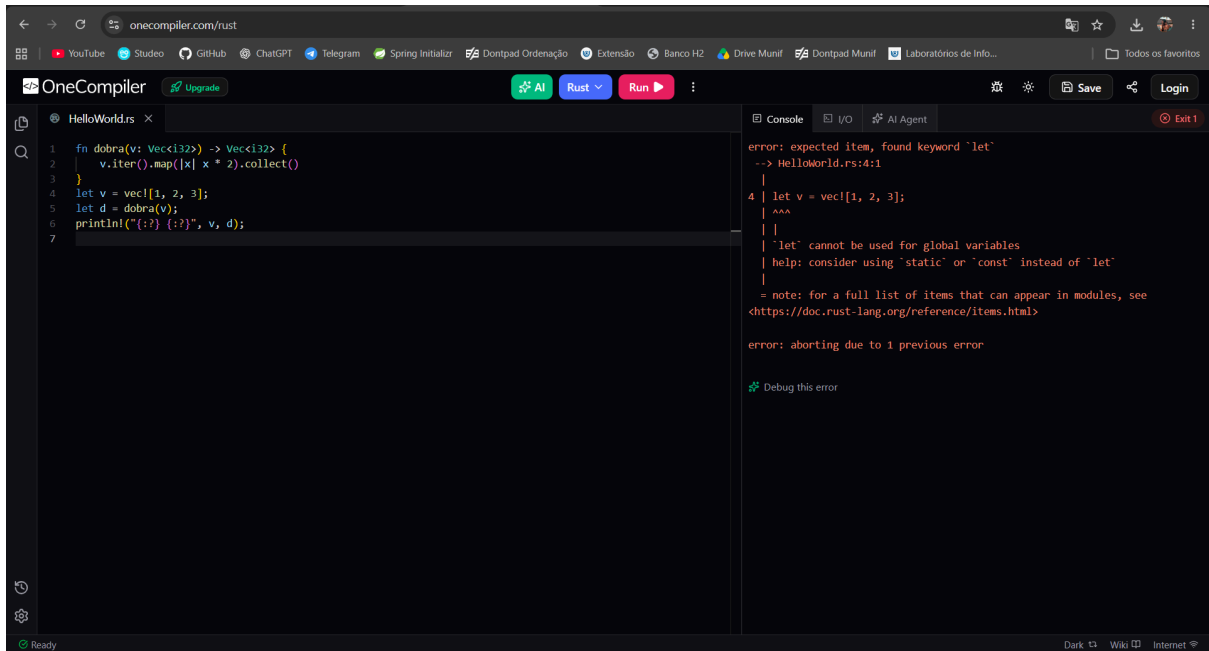


The screenshot shows the OneCompiler C IDE interface. The editor displays a C program in a file named `Main.c`. The code includes `<stdio.h>` and defines a static variable `contador` in a function `contador`. The `main` function calls `contador` twice and prints the result using `printf`.

```
1 #include <stdio.h>
2 int contador(void) {
3     static int n = 0;
4     return ++n;
5 }
6
7
8 int main(){
9     contador(); contador();
10    printf("%d\n", contador());
11 }
```

The console on the right shows the output of the program: `3`. The status bar at the bottom indicates the program is ready and the terminal is active.

Atividade 5 - Rust



The screenshot shows the OneCompiler web IDE interface for Rust. The editor displays a file named `HelloWorld.rs` with the following code:

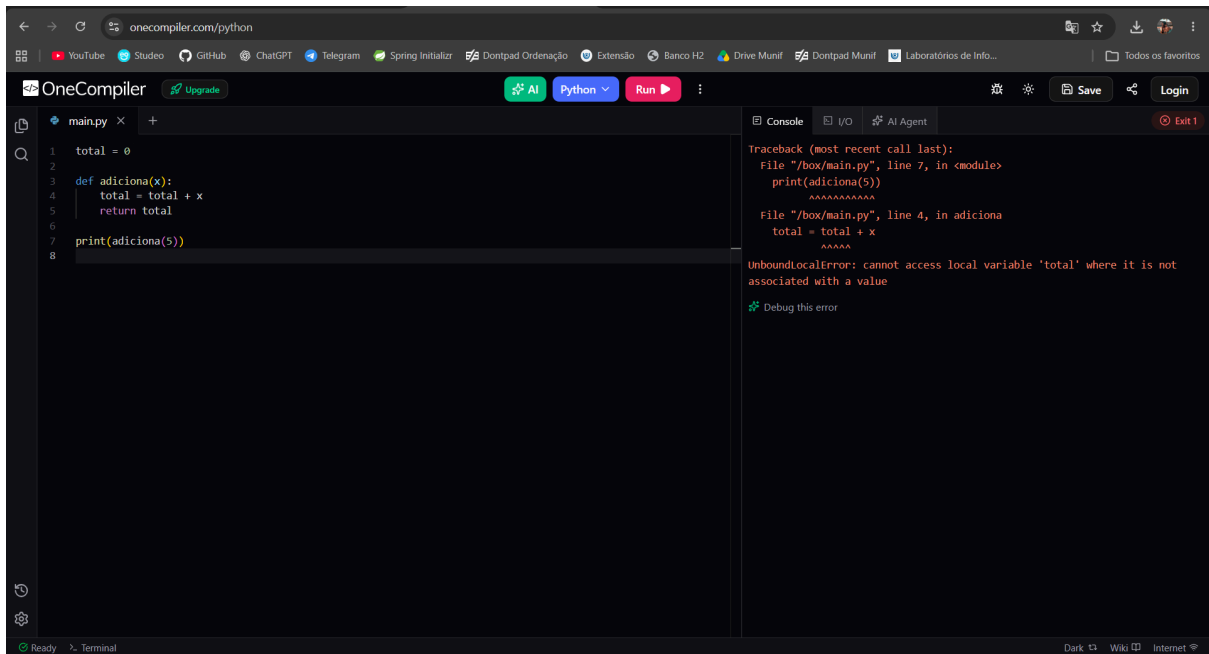
```
1 fn dobra(v: Vec<i32>) -> Vec<i32> {  
2     v.iter().map(|x| x * 2).collect()  
3 }  
4 let v = vec![1, 2, 3];  
5 let d = dobra(v);  
6 println!("{:?} {:?}", v, d);  
7
```

The right-hand pane shows the console output, which contains a Rust compilation error:

```
error: expected item, found keyword `let`  
--> HelloWorld.rs:4:1  
|  
4 | let v = vec![1, 2, 3];  
|   ^^^  
|  
| `let` cannot be used for global variables  
| help: consider using `static` or `const` instead of `let`  
|  
= note: for a full list of items that can appear in modules, see  
       <https://doc.rust-lang.org/reference/items.html>  
  
error: aborting due to 1 previous error
```

Below the error message, there is a link to "Debug this error".

Atividade 6 - Python



The screenshot shows the OneCompiler web IDE interface for Python. The editor displays a file named `main.py` with the following code:

```
1 total = 0  
2  
3 def adiciona(x):  
4     total = total + x  
5     return total  
6  
7 print(adiciona(5))  
8
```

The right-hand pane shows the console output, which contains a Python runtime error:

```
Traceback (most recent call last):  
  File "/box/main.py", line 7, in <module>  
    print(adiciona(5))  
    ^^^^^^^^^^^^^  
  File "/box/main.py", line 4, in adiciona  
    total = total + x  
    ^^^^^  
UnboundLocalError: cannot access local variable 'total' where it is not  
associated with a value
```

Below the error message, there is a link to "Debug this error".